RAJALAKSHMI ENGINEERING COLLEGE

Chandrakanth D 251001039-2025-IT

C2

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|---|---|
| **Started on** | Monday, 14 September 2026, 10:40 PM |
| **State** | Finished |
| **Completed on** | Monday, 14 September 2026, 10:40 PM |
| **Time taken** | 41 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findPair(int arr[], int left, int right, int x)
{
    if (left >= right)
        return 0;

    int sum = arr[left] + arr[right];

    if (sum == x)
    {
        printf("%d\n", arr[left]);
        printf("%d\n", arr[right]);
        return 1;
    }

    if (sum < x)
        return findPair(arr, left + 1, right, x);

    return findPair(arr, left, right - 1, x);
}

int main()
{
    int n, x;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);

    if (!findPair(arr, 0, n - 1, x))
        printf("No");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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